# Marium Zeeshan

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*Computer Science student focused on Generative AI, multimodal systems, and full-stack development. Experienced building scalable solutions from low-level systems to LLM-powered applications, with a strong interest in applied AI research.*

## Education

**Habib University, Karachi** — *2022–2026*
*BS Computer Science* — 100% HUTOPS Scholar
Relevant Coursework: NLP, Data Science, Computer Vision, Digital Image Processing, Deep Learning, Web Development

**Aga Khan Higher Secondary School** — *2019 – 2021*
*Pre-Engineering* — High Achiever: 92.5%

## Technical Skills

**Languages:** Python, C/C++, JavaScript, TypeScript, Verilog
**AI / ML & GenAI:** PyTorch, Transformers, RAG Pipelines, Multimodal Systems (ViLT, BLIP), LangChain
**Web Development:** React.js, Node.js, Express.js, Vite, Tailwind CSS, REST APIs, MongoDB, MySQL, MSSQL
**Data Science:** NumPy, Pandas, Matplotlib, Seaborn, SHAP, EDA
**Tools & Platforms:** Git/GitHub, MATLAB, Vivado, SQL Server Management Studio, WordPress, Docker (basic)

## Experience

**IAL Saatchi & Saatchi (AI Studio)** | AI Intern — *Oct 2026 – Nov 2026*
- Developed a full-stack Influencer Management Platform to manage profiles, filtering, and performance metrics for **5,000+ creators** across Instagram, TikTok, and YouTube.
- Built multi-platform data extraction pipelines using **Apify API** and **Hiker API** to automate creator data ingestion and metadata parsing.
- Implemented ML classification and predictive models to infer creator/audience demographics, geographic locations, and follower trends.
- Structured a normalized **PostgreSQL** database with indexed multi-criteria filtering and metric computation engines.
- Built a responsive frontend dashboard using **React, Vite, and TypeScript** featuring dynamic metric visualization and advanced multi-parameter filtering workflows.

**JDC Foundation** — *June 2026 – Sep 2026*
- Developed a React-based Mortuary Management System handling service bookings, admin approvals, and automated notifications.
- Implemented multi-tier role-based access control (Customer, Admin, Head Admin) with persistent browser-based storage.
- Engineered a bilingual UI supporting real-time English and Urdu script localization for streamlined community operations.

**Optimizers Lab (Pvt.) Ltd.** | Product Development Intern — *Feb 2026 – May 2026*
- Designed and developed responsive websites with improved UI/UX and performance optimisation using HTML, CSS, and JavaScript.
- Implemented server-side logic, database integration, and REST API connections for web applications.
- Managed WordPress themes, plugins, and CI/CD deployment workflows for multiple client projects.

**10Pearls** | Data Science Intern — *Summer 2025*
- Developed a Weather and AQI Prediction System using the Open-Meteo API and Scikit-learn regression models for environmental forecasting.
- Built dynamic dashboards for real-time data visualisation, improving stakeholder accessibility to environmental metrics.

**AlBarkah Bank** | IT Intern — *Summer 2025*
- Engineered a SQL-based database-driven alerting system to monitor service health and system threshold breaches.
- Analysed mobile banking architecture and authentication flows to support secure digital banking operations.

## Publications

**HU at SemEval-2026 Task 10: Psycholinguistic Conspiracy Marker Extraction and Detection** — *Jul 2026*
*Co-author | SemEval 2026 (ACL Anthology)*
- Ranked 1st globally in the SemEval-2026 Task 10 evaluation.
- Implemented Global Pointer architecture with T5, DeBERTa, and GLiNER2 for span-based NER, extracting 5 fine-grained psycholinguistic markers.
- Boosted conspiracy text classification to 0.80 F1 score across imbalanced benchmarks by fine-tuning T5 with custom focal loss and benchmarking against CT-BERT-PRCT and DistilBERT baselines.

## AI & Research Projects

**Multilingual Multimodal Agricultural Chatbot** | Final Year Project (FYP) — *Ongoing*
Industry Collaboration with Pakistan Agriculture Research (PAR)
- Designed a scalable multilingual RAG-based LLM pipeline over 4TB of multimodal data (text, images, audio, video) integrating Qdrant vector search and GPT-4 mini API for real-time agricultural knowledge retrieval.
- Processed multimodal inputs using Whisper Large with VAD for speech transcription, Google Cloud Vision API for OCR, and OpenAI API for English-Urdu translation.
- Optimised context-aware response generation and inference efficiency through advanced prompt engineering and retrieval augmentation techniques.

**Interpretable Anomaly Detection via VLM Fine-Tuning** — *Jan 2026 – May 2026*
- Benchmarked LLaVA, LLaMA 3.2-Vision, and Qwen2.5-VL; fine-tuned 4-bit LoRA on 9,005-image MVTec AD/LOCO benchmark to eliminate output format errors.
- Applied GRPO reinforcement alignment with a custom reward function, reaching 0.93 macro-F1 and 4.3/5.0 LLM-as-a-judge score (Top 3 cohort).

**Visual Question Answering (VQA) System** — *2025*
- Built a multimodal pipeline using ViLT and BLIP architectures to generate natural language answers from visual inputs.

## Technical Projects

**Completely Fair Scheduler (CFS) – OS Kernel Modification**
- Modified the xv6 kernel to implement a Completely Fair Scheduler (CFS) ensuring equitable CPU time distribution across processes.
- Developed a custom `setnice()` system call for dynamic process priority adjustment.

**Cinema Booking System using Red-Black Trees**
- Implemented a cinema seat booking engine using red-black trees for O(log n) data retrieval and seating management.

**Blood Donor App**
- Built a full-stack MERN application for blood donor registration and real-time donation request fulfilment.